Friedrich-Alexander-Universität
Faculty of Engineering

Master’s Thesis

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Philipp Schlatter

Numerical Studies of Transition and Turbulence in a Flat-Plate Boundary Layer

Institute of Fluid Mechanics (LSTM)
LSTM Tech. Rep. 01/2025

Title
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Master's Thesis, Bachelor's Thesis, Project Thesis

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Institute of Fluid Mechanics
Faculty of Engineering
Friedrich–Alexander–University Erlangen–Nürnberg
Prof. Dr. Philipp Schlatter

Author:	NAME
Student Registration Number:	XXXXXXXXXX
Institute:	Institute of Fluid Mechanics
Head of Institute:	Prof. Dr. Techn. Philipp Schlatter
Group and Supervisor:	Prof. Dr.-Ing. habil. Stefan Becker
	Felix Czwielong
Assigned:	dd.mm.yyyy
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List of Symbols

Variable	Einheit	Beschreibung
c_0	m/s	Schallgeschwindigkeit in Luft
∇	$1/m$	Nabla-Operator

List of Abbreviations

Abkürzung	Bedeutung
APE	Aeroacoustic Perturbation Equations
SST	Shear Stress Transport

1 Introduction

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$$G(x) = \sum_{0}^{N} \sin^2(x) \tag{1.1}$$

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1.1 Important stuff

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Figure 1.1: The LSTM logo with the latest colours.

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title 1	title 2	title3
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2 Basics

2.1 DNS

2.2 Turbulent boundary layer theory

- boundary layer develops const in flow direction - with increasing boundary layer thickness - wall shear stress not know a priori - behaviour in inner layer ess. same as in channel flow - equation for pressure gradient - definition of boundary layer thickness, displ. thickness, momentum thickness - corresponding Reynolds numbers - shape factor

2.2.1 pressure gradient

why is it important?

2.3 Optimization??

3 Methods

In the following chapter the methods used in this thesis are presented. In the beginning the used tools are introduced. Then the workflow of the handled project is described and in the end the used optimization method is discussed briefly.

3.1 Tools

To work on this project multiple tools were used. Most of the fluid-flow simulations (korrekte bezeichnung??) were performed with DNS (see XX section 2.1). To perform these simulations the spectral element code NEKO was used. Blabla über neko When running simulaitons with NEKO it is possible to take statistics of the simulated flow during run time. Multiple settings about how the statistics should be recorded can be defined in the case file of the simulation. For example it is possible to define one or two axis along which the statistics should be averaged. These statistics are used to extract and compute certain measures of the flow. To do this some postprocessing of the recorded statistics is necessary. This is done in PYTHON where the package PYSEMTOOLS is included. With this package data recorded in NEKO simulaitons can be extracted and interpolated from an SEM mesh into any arbitrary points (zu nah an github description). Then multiple statistical measures like the budget terms can be calculated. In addition to the DNS, some RANS simulations were performed as well. To run these simulations the open source software OpenFOAM is used. THis software is able to perform RANS simulations of fluid flows with diffenret turbulence models. To extract velocity fields etc from the simulation Python is used. All simulations are performed on the cluster of the RRZE of the FAU. Blabla über cluster

3.2 General workflow

In this section the general workflow followed during this project is presented.

XX Hier noch allgemeines Vorgehesn beschreiben sonst zu detailiert im folgenden XX

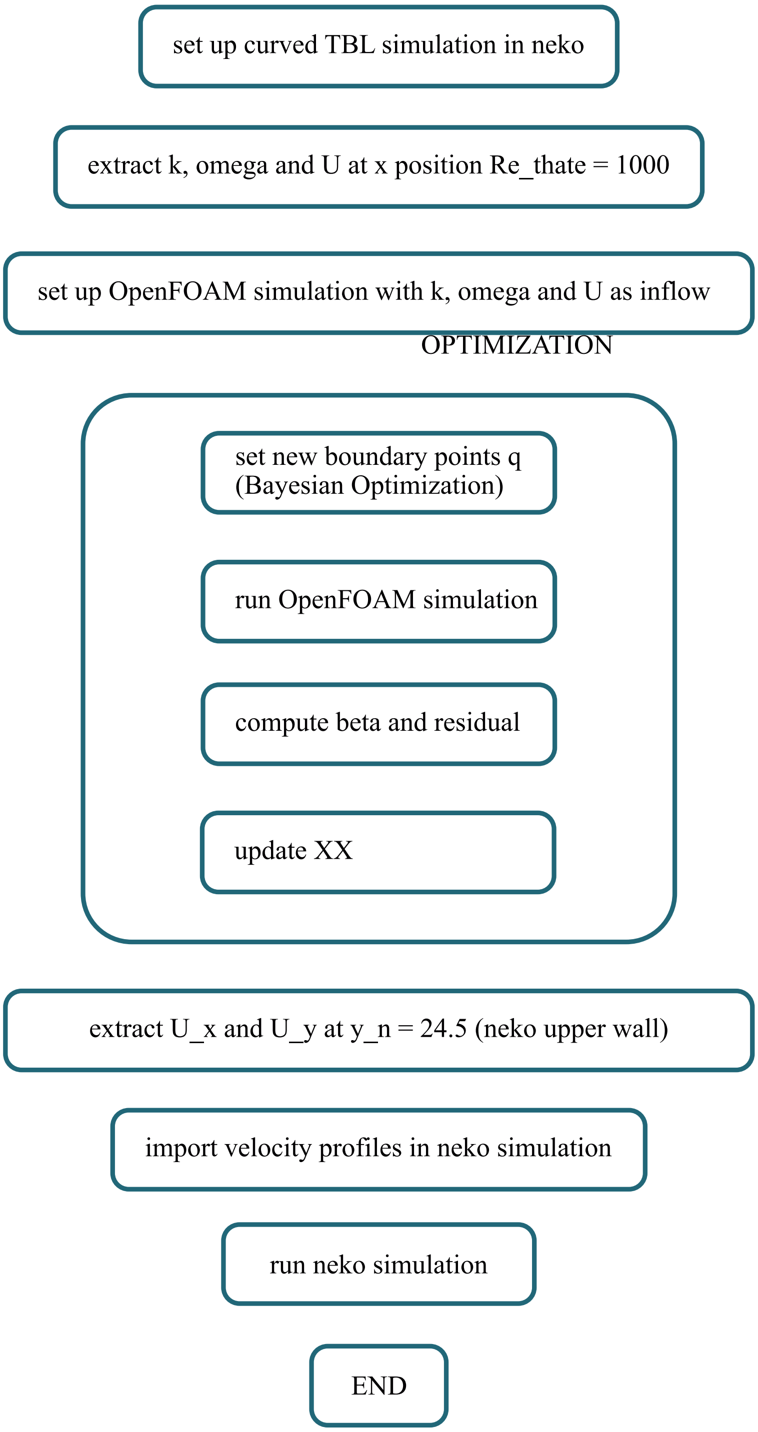

Figure 3.1: Worklow

To get familiar with the NEKO software an example case of the neko example is simulated. We decided to use the channel case since there are some similarities compared to the TBL (see chapter 2). The obtained simulation results as well as the postprocessing procedure are verified and then validated with reference data. In the following step the set up is switched inot a TBL. To do this a new domain with a new mesh is created. At this point a plane TBL is analysed first. A detailed description of the setup can be found in XX. This setup is validated with reference data of other plane TBL simulations eg XX XX. As with the cahnnel simulations, these simulations are used to validate the simulation setup and the postprocessing procedure. IN the final part of the project the curved TBL is analysed. Therefore it is necessary to build a simulation setup where a ZPG is ensured such that only curvature effects can be analysed. To create such a setup multiple steps are necessary which are depicted in Figure 3.1. First a working setup of a curved TBL without pressure modification is established in NEKO including the domain size, case settings, vector rotations etc. which are descriped in detail at XX. With this setup a simulation is performed. The results of this simulation are used as inflow conditino for the following RANS simulations in OpenFOAM. More precisely the measure of U, k and omega at the positoin of $Re_theta = 1000$are extracted. Then the OpenFOAM simulation is set up such that the profile at $Re_theta = 1000$ from the DNS is used as boundary condition at the inflow of the domain in the RANS simulation. XX hier noch das OPenfoma domain höher ist XX With this simulation setup the optimization process is started. The first step of the optimization loop is to ste new boundary points q. In the beginning (if there are less then XX in GPinit) the points are taken randomly in the given parameter range. in the next step. The domiain is build with the new boundary points and the OpenFOAM simulation is started. When the simulation is finished, the postprocessing is performed to compute the current Clauser parameter β and the residual R. As a last step the files and XX are updated. If the optimization was performed for a certain number of steps the new boundary points q are selected with the bayesian optmization. <the optimization runs until ???XX, sich that the ZPG is ensured. In the next step the velocity profile for the optimized domain is extracted. The values are extracted from the RANS simulation at the wand normal position which corresponds to the topwall of the DNS domain. This velocity profile is imposed to the DNS setup and then the DNS simulation is performed.

- einarbeiten in neko software –> channel simulations - perform simulations and verify results with reference data
- create TBL domain - perform plane tbl isimulations and validate the results with reference data
- create curved domain - set uo neko settings for working simulaiton - copy setup in OpenFOAM
- run optimization
- beta (clauser parameter) at the lower wall as target measures - odjective function: distnace between beta at the bewertungspoints q nad prescribed beta along the lower wall (at the same x position)
- extract results (U, ??) from OpenFOAM - impose these data as boundary condition at top wall in neko simulation - perform neko simulation
- do this for different domains: why these domains?

3.3 Optimization Methods

Bibliography

P. Schlatter and R. Örlü. Assessment of direct numerical simulation data of turbulent boundary layers. *Journal of Fluid Mechanics*, 659:116–126, 2010.

A Einleitung

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